Final Project Paper



ECON3253: Research Methodology (WI)

Online Women Entrepreneurs during COVID'19 Pandemic:

A study on Chittagong City, Bangladesh

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1. Abstract

Although COVID'19 has badly impacted the whole world, one positive impact can be the increase in use of technology. Use of technology became a part of people's life, by making study at home and work from home easier. One of the successes of technology can be Online business which is playing a role in increasing the employment rate in different countries, like Pakistan and Canada. However, women of Bangladesh took the chance and started their own business. Literature from different countries say that women through training and practice are trying to keep up their online business and so online business plays an important role in increasing women employment during the pandemic. This study looked into the factors that are influencing the women entrepreneurs in Bangladesh to carry on their online business and if it is feasible to continue after COVID. Different analysis methods like, univariate analysis, bivariate analysis and softwares like STATA, Google Sheet were used to find the result from 50 women entrepreneurs of Chittagong City. The result shows that the majority of women started their business during the COVID. The Cloth & Jewellery businesses are the most prominent business among the online businesses. Majority of women entrepreneurs have a housemaid and they get support from their family. After being asked the question of continuing business after COVID, the majority replied that they want to continue their business and it is possible to continue the business.

Keywords: *Entrepreneurs, Online Business, Univariate Analysis, Bivariate Analysis, Multivariate Analysis*

2. Introduction

Data from the World Bank shows that the unemployment rate in Bangladesh in 2019 (pre-COVID'19 period) was 4.20 percent and the unemployment rate in Bangladesh in 2020 (COVID'19 period) was 5.30 percent (World Bank, "Unemployment"). From 2018 to 2019, the unemployment rate declined by 0.07 percent, while it increased by 1.08 percent from 2019 to 2020. The COVID'19 pandemic caused a rise in unemployment rate which is one of the most important economic crises in Bangladesh. Unemployment is very harmful for a country's development as it increases the poverty rate. Unemployment has always been a very important issue for developing countries like Bangladesh. COVID'19 pandemic in this case added fuel to

the fire called unemployment. However, there is one sector which started increasing during the COVID- the business sector - more specifically, the *Online Business* sector.

It is necessary to come up with an alternative to solve the job crisis and one of the alternatives can be working on online platforms. The whole population of Bangladesh can not work online due to lack of technology availability and lack of training on use of technology. A portion of the population, like the people who are interested in doing business and know the use of technology, can use online platforms to earn money. If the business can not be handled alone, the business-owner can also employ a few more employees to maintain the business. Moreover, studies found in different countries show that online business creates opportunities for women entrepreneurs and this can help increase women employment rate in Bangladesh.

This paper attempts to identify the impact of digital technology in creating new employment opportunities in Bangladesh focusing on online businesses and rising entrepreneurs during the COVID'19 pandemic. Thus the paper will look to answer the following question:

- How are online businesses influencing new women entrepreneurs in Chittagong during the COVID'19 pandemic?

This paper will also look at the success or failure of the new women entrepreneurs that might affect the continuation of their businesses. Thus, the following question rises:

- Is online business effective enough to continue after COVID'19?

3. Objectives of this Study:

The paper analyses the factors affecting the decisions of women entrepreneurs who are operating their businesses in online platforms during the COVID'19 pandemic and if it is possible to continue their business after the pandemic. More specifically, the objectives of this study are:

- ➢ Factors influencing the Online Business
- ➢ Factors influencing the Continuity Decision
- ➢ Possibility to continue after COVID

4. Literature review:

The COVID'19 Pandemic, and the public health interventions taken to control it, created widespread social disruption resulting in harming studies of children as schools were closed,

shuttered business, rise in unemployment, increase of emotional distress and increase of mental health issues (Garfin, 2020). Social support is an important factor that can help people relieve stress and digital technology like social media has a powerful impact on people in helping them to reduce anxiety (Drouin, McDaniel, Pater & Toscos, 2020). As time passed by, people started to accept the "new normal" situation, i.e., permanently normalizing the use of digital technology in education, business, healthcare and social networking. Thus, students and employees through online classes and video-conferencing meetings made studying from home and work from home a new daily life routine (Herath & Herath, 2020).

Digital technology has enabled e-learning and e-commerce for people around the world. Digital technology has made life easier by motivating people to learn anything they want or work by themselves by starting a business online (Apăvăloaie, 2014). Experts during 2006 predicted that in the future, the internet will play an important role in business and become a primary low-cost network for transactions. The producers and consumers will benefit from online purchasing by saving costs and improving productivity (Deeter‐Schmelz, Bizzari, Graham & Howdyshell, 2006). The prediction became true during the COVID'19 pandemic when buyers to maintain social distancing, started online purchasing of food, clothes, medicine and daily necessities, which gave rise to sellers of the commodities that consumers demand for. Buyers have specific expectations on the quality and service of the goods that they purchase online and if the expectations are met, sellers can make a profit out of it (Neger & Uddin, 2020).

In the case of online business, social media plays an important role. Small and medium-sized enterprises showcase their products on the company's website and share the link with friends and family on social media (Ahmad, Ahmad & Bakar, 2018). Some small businesses entrepreneurs use social media platforms for advertising by creating fan pages, publicizing their products and selling their products to the buyer willing to buy. Satisfying the buyers by providing products of good quality, the entrepreneurs can build a long term relationship with the customers and can continue with the business for a long time (Shabbir, Ghazi & Mehmood, 2016). However, online businesses are more preferred by women entrepreneurs as women have the responsibility of households, and because it is online, they can take care of business and household works at the same time. Online businesses are increasing opportunities for women entrepreneurs; looking at the benefits of online business and noticing the opportunities available in social media motivates them to open their own business (Hossain & Rahman, 2018).

Although women play a significant role in the economy of Canada, studies found that women entrepreneurs are small scale business owners, for which they lack opportunities and have to face financial stress (Grandy, Cukier & Gagnon, 2020). Studies found that women entrepreneurs are always in a disadvantageous situation because of being the minority group in a male-dominating society, and COVID'19 pandemic has affected the women business owners in the UK heavily (Martinez & Jayawarna, 2020). A study by Jaim (2021) found that the COVID'19 pandemic has made women business owners in Bangladesh close their business as it is difficult for women entrepreneurs to take care of business, household works and child care at the same time. Studies found on Pakistani women business owners show that women do not only face financial crisis due to decrease in business sales, but also face challenges in household income, lifestyle and mental stress (Mustafa, Khursheed, Fatima & Rao, 2021). However, none of these studies mentioned the case of women entrepreneurs who have been operating their businesses online.

A study found that COVID'19 pandemic opened an opportunity for women business owners; women business owners who were suffering from supply and demand shock due to the pandemic pivoted their business model from manual business to online business and started experiencing rise in supply and demand (Manolova, Brush, Edelman & Elam, 2020). A study found that online business did not only open opportunities for existing women entrepreneurs, but also for new women entrepreneurs in developing countries after they did online courses on improved business strategies, and learned: developed strategies of producing final products from raw materials, making it available and attracting the customers, delivery on time and managing the online transactions (Afshan, Shahid & Tunio, 2021). Another study on the Khyber Pakhtunkhwa community of Pakistan says that COVID'19 pandemic has been a blessing for cloth and cosmetics women business owners but the case is different for women entrepreneurs having business in dairy products (Muhammad, Ximei, Haq, Ali & Beutell, 2021).

A small volume of literature on online business during the COVID'19 has been published. So far as best of my knowledge, no literature has been published discussing the Bangladeshi women entrepreneurs who have been operating their business on an online platform. Online business has increased opportunities for women entrepreneurs in Bangladesh, but how are the existing business women continuing their business and what is influencing the new women entrepreneurs to start their business during the pandemic situation are important questions to

know. This paper will provide a simple theoretical framework on the factors influencing the women entrepreneurs of Chittagong city, Bangladesh to operate online business during the COVID'19 pandemic. This paper will also try to find out if online business is effective enough to continue after COVID'19 or if the entrepreneurs are willing to continue their online business after COVID'19 pandemic.

5. Data

This study is a Quantitative study. The sampling method used to select the respondents is Convenience Sampling. Online business owners whom I follow on Facebook are being selected for this study. Due to COVID'19 pandemic it is not possible for me to use a random sampling method. Also it is difficult to get a list of all online women business owners and thus could not be included in my sampling frame. So the business women who are not on my facebook follow list are not selected for this study, who may have a different answer from which are collected for this study. Thus the findings of the paper can not generalize the overall women entrepreneurs of Chittagong City.

In this research primary data is collected from 50 samples of women entrepreneurs living in Chittagong City of Bangladesh. It is to be noted that all 50 respondents that are targeted for this study are operating online business from Chittagong City and so can not be a representative of the overall population of Bangladesh. This paper has a relational survey design, and thus sample size 50 is somewhat a good sample. 4 varieties of women entrepreneurs participated in this sample: married, single, divorcee and single parent. As the population of women entrepreneurs is homogeneous, the variation is less and a smaller sample can be used to reflect the varied population. Besides, my sample size is constrained to time and cost. The response rate for this study is 89.2%, which can be called a good response rate.

All the data used are collected by administering a self completion questionnaire to all the 50 participants, that is, the participants were asked to fill up questionnaires in Google form and submit through email, messenger and whatsapp. In the case of forming questionnaires, I have used both close ended and open ended questions. I could not personally interview all the participants and so probing and prompting in this study were absent. In order to reduce biases from the study, most of the questions are close-ended. However, not all questions are

close-ended, to collect the valuable information I kept two open ended questions, so that respondents can share their ideas.

The study has individual level information on the factors influencing the women entrepreneurs to operate online business and if it is possible for them to continue business after COVID'19.

$$[* \text{ Response Rate} = \frac{Number\ of\ Usable\ questionnaires}{Total\ Sample - Uncontactable\ Members\ of\ the\ Sample} \text{ X } 100$$

$$= \frac{50}{68 - 12} \text{ X } 100 \ = 89.2\ \% \ ]$$

6. Methodology:

After all the data collected, the answers were coded to find out what factors are influencing the surveyed women entrepreneurs and if the women are willing to continue their business after COVID. By analyzing the relation between different factors like, business type, housemaid's help, family's support and women entrepreneurs' continuity decision, I tried to conclude by using multivariate regression analysis, univariate analysis and bivariate analysis. Google Sheet and STATA software have been used to process survey data.

6.1 Types of Variables

The questionnaires used for this study contain a series of Open ended questions, Nominal questions, Dichotomous questions, Interval questions and Ordinal questions. *Table 1* shows the list of variables of my study as per their type of variables (Interval, Ordinal, Nominal, Dichotomous).

Table 1: List of Variables and Variables of this study

Type	Example from Women Entrepreneur study	Variable Name
Interval	var0005	Age
	var0007	Household work hours
	var0008	Business work hours
	var0018	Investment (BDT)
	var0019	Profit (BDT)

Ordinal	var0014 var0015 var0016 var0017 var0020	Experience of business Customers Responses from customers Profit satisfaction Comparison
Nominal	var0002 var0003 var0004 var0006 var0011 var0012 var0013 var0021 var0022	Business type Starting time (pre/post COVID'19) Start of Operation Household works Marital Status Family Support Help from Family Continuing after Covid'19 Possibility of continuing after Covid
Dichotomous	var0001 var0009 var0010	Online business operator Availability of Housemaid Housemaid type

[var0020 and var0022 are open ended questions. In var0020, the answers collected can be arranged as range, for instance, "Online Business is easier or moderate or tougher than Traditional Business", here the difference between easier and moderate and tougher is not equal. In var0022, the answers collected comprise categories like "yes", "no" and "maybe" that can not be rank ordered.]

6.2 Data Analysis Technique

Univariate Analysis, *Bivariate Analysis* and *Regression Analysis* done for this study are mentioned in this part. According to the objectives of my study, the analysis techniques used for each variable mentioned in *Table 1* are divided and discussed in the following few sections. To show the variation of the background of my respondents, I used demographic variables like Var0005 (Age) and Var0011 (Marital Status) and showed them in percentage form in a frequency table. These two variables are not used directly to answer my research questions.

6.2.1 Analysis for Factors Influencing Online Business

From previous literature, it is seen that there are some factors which are influencing online women entrepreneurs to operate business online during the COVID'19. Some factors include 1. COVID 2. Type of business 3. Availability of housemaid and 4. Family's support. I used both univariate and bivariate analysis to answer this part.

(A) Univariate Analysis

For univariate analysis on what factors are influencing the women entrepreneurs in Chittagong city, I will be using frequency tables, bar charts, pie charts and boxplot. *Table 2* below gives the list of variables used for univariate analysis.

*Table 2***:** Univariate Analysis of Influence on Online Business

Frequency table	Var0002 (Nominal) Var0006 (Nominal) Var0012 (Nominal)	Type of Business Household work Family Support
Histogram	var0007 (Interval) var0008 (Intervall)	Household working hours Business working hours
Pie Chart	var0013 (Nominal)	Help from Family
Boxplot	var0018 (Interval) var0019 (Interval)	Investment Profit

(B) Bivariate Analysis:

For bivariate analysis, I will be using a contingency table, Pearson's r and Scatter Diagram to show the correlation between different variables. *Table 3* below shows the list of variables used in bivariate analysis.

Table 3: Bivariate Analysis of Influence on Online Business

Contingency Table	var0003 (Nominal) + var0004 (Nominal) var0009 (Dichotomous) + var0010 (Dichotomous)	Starting time & Starting of Business Availability of Housemaid & Housemaid Type
Pearson's r & Scatter Diagram	var0019 (Interval) + var0008 (Interval)	Profit Business working hours

6.2.2 Analysis for Factors Influencing Continuity Decisions

To analyze this part, I used a specific question: *Do you want to continue your online business after COVID'19?* This question is var0021 mentioned before, and I used this variable by the name *Continuity Decision.* This particular variable is used in all analyses that are done for this part. I used the factors that are influencing online business to see if the factors further influence the women entrepreneurs to continue their business. The factor COVID is not used, as the continuity decision is after COVID and so this factor does not have a role to play in this part. I used bivariate analysis and regression analysis for this part.

(A) Bivariate Analysis

For bivariate analysis, I will be using a contingency table and Cramer's V correlation method. *Table 4* below shows the list of variables used in bivariate analysis.

Table 4: Bivariate Analysis for Factors Influencing Continuity Decision

Contingency Table	var0021 (Nominal) + var0019 (Interval)	Continuity Decision & Profit
Cramer's V	var0021 (Nominal) + var0002 (Nominal) + var0009 (Dichotomous) + var0012 (Nominal)	Continuity Decision, Business type, Availability of Housemaid Family Support

(B) Regression Analysis

Taking continuity decision as dependent variable and the four factors as independent variables with error term ε, we can get the following online business women model,

$$\textbf{\textit{Continuity decision}} = \beta_1 + \beta_2\, \textbf{\textit{BusiType}} + \beta_3\, \textbf{\textit{HM}} + \beta_4\, \textbf{\textit{FamSup}} + \varepsilon_{Cd}$$

where, BusiType, HM and FamSup represents type of business, availability of housemaid and family's support respectively, and ε_{Cd} is the error term.

Interpretation of β_1*:* Considering, type of business, availability of housemaid and Family's support has no effect on online business, (i.e. BusiType = HM = FamSup = 0), the decision of online women entrepreneurs to continue their business after COVID'19 pandemic will change by β_1 units.

Interpretation of β_2*:* Considering, availability of housemaid and family's support has no effect on online business, (i.e. HM = FamSup = 0), if entrepreneurs come up with 1 more type of

business, the decision of online women entrepreneurs to continue their business after COVID'19 pandemic will change by β_2 units.

Interpretation of β_3*:* Considering, type of business and family's support has no effect on online business, (i.e. BusiType = FamSup = 0), if availability of housemaid in 1 businesswoman owner's house increases, the decision of online women entrepreneurs to continue their business after COVID'19 pandemic will change by β_3 units.

Interpretation of β_4*:* Considering, type of business and availability of housemaid has no effect on online business, (i.e.BusiType = HM = 0), if family member's support for 1 online businesswoman increases, the decision of online women entrepreneurs to continue their business after COVID'19 pandemic will change by β_4 units.

I used the overall significance test F-test to find out if my model is statistically significant. Furthermore, I showed confidence interval, r-square value and heteroscedasticity of the model to show if β_1, β_2, β_3 and β_4 are linearly dependent on my dependent variable *Continuity Decision*.

6.2.3 Analysis for Possibility to Continue after COVID

To find out if online business is effective enough to continue after COVID'19, few questions were asked to the participants, for instance, (i) *What do you think about Online business in comparison to Traditional business?* and (ii) *Is it possible to continue online business after COVID'19? Will the profit and response from customers be the same?*. These questions are var0020 and var0022 that are mentioned before respectively. Both of these questions are open-ended questions. However, due to the constant answers received from the survey, I was able to divide them into different variables. To analyse this part, I used univariate analysis and bivariate analysis.

(A) Univariate Analysis

Table 5 shows the list of variables that are used for univariate analysis.

Table 5: Univariate Analysis for Possibility to Continue

Frequency Table	var0020 (Ordinal) var0022 (Nominal) var0014 (Ordinal) var0015 (Ordinal)	Comparison Possibility to Continue Experience of business Customers

	var0016 (Ordinal) var0017 (Ordinal)	Responses from customers Profit satisfaction

(B) Bivariate Analysis

For bivariate analysis, I will be using spearman's rho to show the correlation between different variables. *Table 6* below shows the list of variables used in bivariate analysis.

Table 5: Bivariate Analysis on Continuation of Online Business

Spearman's rho	var0014 (Ordinal) + var0015 (Ordinal) + var0016 (Ordinal) + var0017 (Ordinal)

7. Findings & Analysis:

To find out the factors influencing women entrepreneurs in Chittagong city and decision about continuing their online business after COVID, findings and analysis are presented below based on the questions of the questionnaire used in this study.

7.1 Using Univariate Analysis

At the very beginning of my data collection, I have classified the respondents by their age, type of business, household works, family support and marital status according to their responses.

Table 6 : Demographic Characteristics of Respondents (Frequency Distribution)

Demographic Factors	No. of Respondents	Percentage (%)
Age		
16 - 25 years	18	36 %
26 -.35 years	15	30 %
36 - 45 years	13	26 %
46 - 55 years	4	8 %
Type of Business		
Cloth & Jewelleries	25	50 %

Bakery	11	22 %
Henna	3	6 %
Cosmetics	5	10 %
Graphic Design	1	2 %
Organic Food	5	10 %
Household Works		
Yes	19	38 %
No	24	48 %
Sometimes	7	14 %
Family Support		
Yes	32	64 %
No	12	24 %
Sometimes	6	12 %
Marital Status		
Married	24	48 %
Single	21	42 %
Divorcee	3	6 %
Single Parent	2	4 %
Comparison (Online Business or Traditional Business)		
Online Business is Easier to handle & saves time	34	68 %
Online Business is tougher to handle & time consuming	16	32 %

[* STATA command for Frequency table: *tab variable_name*]

The frequency distributions in *Table 6* shows that the age range of the majority (36%) of women entrepreneurs among my respondents were from the 16 to 25 years age group out of 100%, 30% were from 26 - 35 years, 26% from 36 - 45 years and only 8 % were in age group of 46 to 55 years. The table highlighted that 50% of the respondents are operating Cloth &

Jewelleries' business online, which is the majority. As for other businesses, bakery business is operated by 22%, 10% has cosmetics business, another 10% are operating organic food business, henna business is conducted by 6 % and only 2% are doing business related to graphic design.

In the table above, it can be seen that 38% of women entrepreneurs among my respondents have to do housework, 48% do not have to work and 14 % work sometimes. 64 % of online entrepreneurs get family support and help from husband, parents or siblings, while 24 % do not get any support and 12 % sometimes get support. In this study, 48 % women are married, 42 % are single, 6 % are divorcees and only 4 % are single parents operating online businesses. The table shows that 68 % of respondents answered, online business is easier to handle and saves time in comparison to traditional business. 32 % of respondents say that online business is tougher to handle & time consuming.

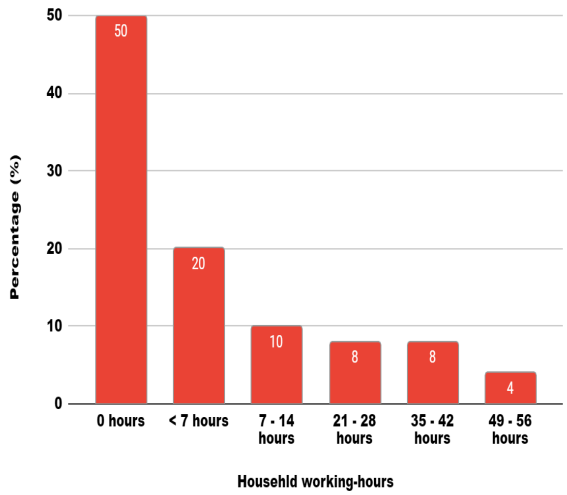


Figure 1: Household Working-hours(%)

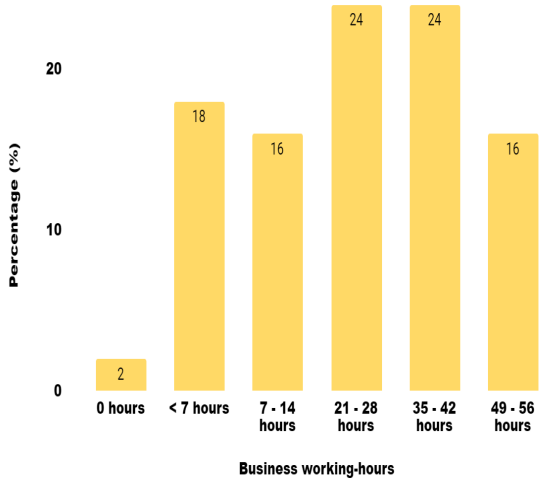


Figure 2: Business Working-hours(%)

The bar chart in *Figure 1* indicates the hours women business owners spent in doing household chores. 50 % of business owners out of 100% in this study do not do any household chores, because of the availability of housemaids, which is the majority. 20 % of my respondents spent less than 7 hours per week on household chores. 10% spent 7 to 14 hours per week, 8% spent 21 to 28 hours per week and another 8% spent 35 to 42 hours per week doing household chores. Only 4% spent 49 hours to 56 hours on household chores per week.

The bar chart in *Figure 2* highlights the hours women business owners spent in their business works. 24% of business owners out of 100% in this study spent 21 to 28 hours per

week and another 24% spent 35 to 42 hours per week doing business work. 18% spent less than 7 hours per week, 16% spent 7 to 14 hours per week and another 16% spent 49 hours to 56 hours on business work per week. 2%, that is one respondent closed her business recently and so does not spend any time on business work.

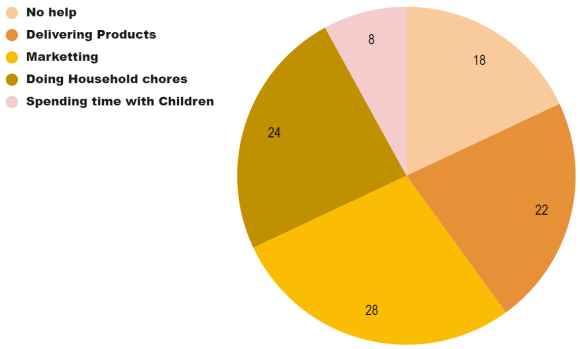


Figure 3: Help from Family in Percentage (%)

Pie chart in *Figure 3* shows the percentage and type of help the women business owners receive from their families. The majority of respondents (28%) answered that their family members help them with marketing their products. 24% of the respondents' family helps them by doing the household chores leaving them with free time to take care of the customers. 22% of online business owner's family members help them with delivering the products. 8% of the married respondents can spend more time in business work because their husband helps them with taking care of the children. 18% of the business owners do not receive any help from family members.

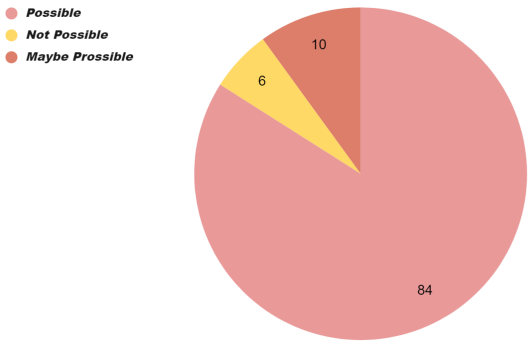
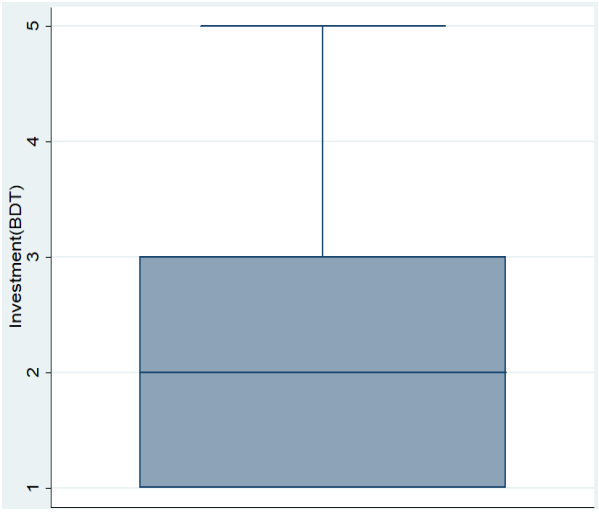
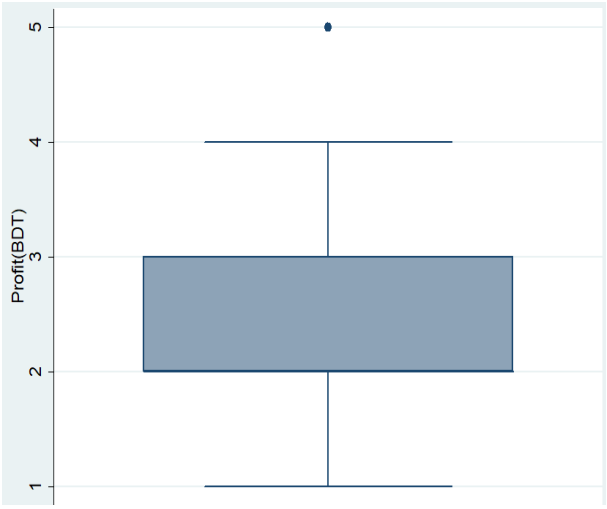


Figure 4: Possible to Continue Online Business After COVID in percentage(%)

Pie chart in *Figure 4* shows that 84% of my respondents agree that it is possible to continue their online business after COVID. 6% of the online business owners want to shift their online business to manual business after COVID. 10% of the respondents are not sure about possibility of continuation after COVID; these business owners started business online and have no idea about manual business. According to them, the possibility depends on their customers, if the customers' responses remain the same then only they will continue.





[*STATA command for Boxplot: *graph box variable_name*]

Figure 5: Investment (BDT)

Figure 6: Profit (BDT)

A Box plot is an effective visual representation of both central tendency (the median) and dispersion (the range). Box Plot in *Figure 5* shows the Investment of online business owners to start their business in BDT. The maximum range of Investment is BDT 40,001 to BDT 50,000 and the minimum range is less than BDT 10,000. 50% of the respondents in this study invested up to BDT 30,000 to start their business. Here the box lies in the lower end, that means the highest variability is towards the lower value of the distribution. The median ranges from BDT 10,001 to BDT 20,000 for investment. There are no outliers in this distribution.

Box Plot in *Figure 6* shows the Profit that the online business owners earn every month in BDT. The maximum range of Profit is BDT 30,001 to BDT 40,000 and the minimum range is less than BDT 10,000. 50% of the respondents in this study earn a profit between BDT 10,001 to

BDT 30,000 every month from their business. As the box lies in the middle, that means the highest variation is in the middle. The median ranges from BDT 10,001 to BDT 20,000 for Profit. Here, there is one outlier, which contains the range BDT 40,001 to BDT 50,000.

Table 7: Business Ratings

	5	4	3	2	1
Experience of business	22 %	54 %	22 %	2 %	0 %
Customers	22 %	52 %	26 %	0 %	0 %
Responses from customers	50 %	28 %	22 %	0 %	0 %
Profit satisfaction	42 %	34 %	22 %	2 %	0 %

[* STATA command for Frequency table: *tab variable_name*]

Ratings on Business indicated in *Table 7* were collected from the respondents in scale of (1-5). The ratings were taken under four factors, as discussed below:

Experience of Business: This was rated in the scale of Poor to Excellent (1 to 5). Majority of the respondents (54%) rated 4, which can be said to be a 'good' experience. 22 % of the respondents rated 5, which is 'excellent' experience and another 22% rated 3, which is 'satisfactory' experience. 2 % rated 'moderately poor' experience, which is only 1 respondent who has stopped her business during the COVID. No respondents rated scale 1.

Customers: This was rated in the scale of Rarely to Frequently (1 to 5). Majority of respondents (52 %) rated 4, which can be said that 'usually' get customers. 26 % rated 3, which can be said 'normally' get customers and 22% rated 5, which is 'frequently' get customers. No respondents rated scale 1 and 2.

Responses from Customers: This was rated on the scale of 'Customers dislike my Products' to 'Customers like my Product' (1 to 5). Majority of respondents (50%) rated 5, which means their customers like their products. 28% of respondents rated 4, which means customers moderately like their products. 22% rated 3, which can be said their customers are satisfied with the products. No respondents rated scale 1 and 2.

Profit Satisfaction: This was rated in scale of 'Not Satisfied' to 'Very Satisfied' (1 to 5). Majority of respondents (42%) rated 5, which means these business owners are 'very satisfied' with the profit they are earning from online business. 34% rated scale 4, which means the business owners are 'moderately satisfied' with the profit they are earning. 22% of the respondents rated 3, which is they are 'satisfied' with their profit. 2 % of the respondents, which is only 1 respondent rated scale 2, which means she is 'moderately not satisfied' with the profit she was earning and so she stopped her business. No respondents rated scale 1.

7.2 Using Bivariate Analysis

For bivariate analysis, I classified my respondents according to business starting time (pre/post COVID'19), availability of housemaid and decision of continuation after COVID.

Table 8: Demographic Characteristics of Respondents(Contingency table)

Starting Time

	Started Manually, sifted to online	**Started Online, continuing online**	**Started online**	**Total**	**(%)**
Before COVID	14 %	26 %	0 %	20	40 %
During COVID	4%	0 %	56 %	30	60 %

Availability of Housemaid

		Full-timer	**Part-timer**	**Total**	**(%)**
Has Househelp	0 %	24 %	42%	33	66 %
No Househelp	34 %	0 %	0 %	17	34 %

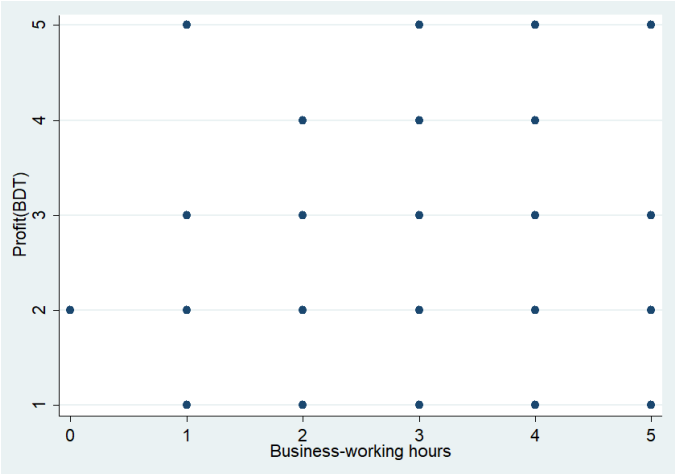
Continuing after Covid'19

Profit/ Decision	**< BDT 10,000**	**BDT 10001 -BDT 20000**	**BDT 20001 -BDT 30000**	**BDT 30001 -BDT 40000**	**BDT 40001 -BDT 50000**	**Total**	**(%)**
Yes	20%	24%	18%	6%	10%	39	78 %
No	0%	8%	0%	0%	0%	4	8 %
Maybe	2%	6%	6%	0%	0%	7	14 %
Total	22%	38%	24%	6%	10%	50	100%

[* STATA command for Contingency Table: *tab variable1_name variable2_name***]**

It is clear from Table *8* that 40% of my respondents started their business before COVID; among them 14% entrepreneurs started business manually and shifted to online business due to COVID and 26% entrepreneurs started online before COVID and are continuing online during COVID. However, the majority, 60% started business during COVID; among them 56% started business online, 4% started business manually and later shifted to online. From Table, 66 % of my respondents have househelp, among which 42% are part-timer and 24% are full-timer; 34 % do not have any househelp.

Moreover, answering the question of if they want to continue their online business after COVID lockdown, 78% confirmed that they will continue their business which is the majority, among them, 20% entrepreneurs are earning profit less than BDT 10,000, 24% are having profit of BDT 10,001 to BDT 20,000, 18% are earning profit of BDT 20,001 to BDT 30,000, 6% entrepreneurs earn profit of BDT 30,001 to BDT 40,000 and 10% entrepreneurs are earning profit between BDT 40001 to BDT 50000. 8% of the current online business women do not want to continue their business after COVID, and all of them are earning a profit of BDT 10,001 to BDT 20,000. 14% of my respondents are unsure about continuing their business after COVID, among them 2% entrepreneurs earns profit less than BDT 10,000, 6% are having profit of BDT 10,001 to BDT 20,000 and 6% entrepreneurs earn profit of BDT 20,001 to BDT 30,000.



[* STATA command Scatter Diagram: *scatter profit businessworkinghours*]

Figure 7: Correlation between Profit and Business-working hours

Scatter diagram in *Figure 7,* Profit has only 5 values and Business-working hours also has only 5 values. The trend in the scatter diagram shows that there is a very weak relation between both the variables, that is the correlation will be closer to 0. In the *Table 9* below, Pearson's r method is used to find the correlation coefficient between the two interval variables and the correlation coefficient is 0.114. It can be said that not only the variation in Profit from online business is influenced by Business-working hours but also there are more other variables that are influencing the relationship.

Table 9 and *Table 10* below shows the correlation between different variables.

[***Note***:

- Correlation Coefficient between 0 to ± 0.2 is considered Very Weak correlation
- Correlation Coefficient between ± 0.2 to ± 0.4 is considered Weak correlation
- Correlation Coefficient between ± 0.4 to ±0.7 is considered Moderate correlation
- Correlation Coefficient between ±0.7 to ±1 is considered Strong correlation]

Table 9: Factors influencing the Continuity Decision

	Continuity decision	***Business Type***	***Availability of Housemaid***	***Family Support***	***Profit***	***Business working-hours***
Continuity decision	1.000					
Business Type	0.263	1.000				
Availability of Housemaid	0.364	0.321	1.000			
Family Support	0.146	0.383	0.126	1.000		
Profit					*1.000	
Business working-hours					*0.114	*1.000

[* indicates that Pearson's r is used**]**

[* STATA command for Cramer's V : *tab variable1_name variable2_name, V*]
[* STATA command for Pearson's r: *corr variable1_name variable2_name*]

Cramer's V method used in *Table 9* gives the correlation between different factors that are influencing Online business and the decision of continuity of online business after COVID.

- The correlation coefficient between *Continuity Decision* and *Business Type* is *0.263,* that is greater than 0.2, means that there is a weak positive relationship between both the variables. Continuity decisions are influenced by some other factors other than business type.
- The correlation coefficient between *Continuity Decision* and *Availability of housemaid* is *0.364*, means that there is a weak positive relationship between both the variables but continuity decisions are influenced by some other factors other than availability of housemaid.
- The correlation coefficient between *Continuity Decision* and *Family Support* is *0.146*, that is less than 0.2, means that there is a very weak positive relationship between both the variables and continuity decisions are influenced by some other factors other than family support.
- Besides, the correlation coefficient between *Business Type* and *Availability of housemaid* is *0.321*, and that of *Business Type* and *Family Support* is *0.383*. This indicates that the *Business Type* has a weak positive relationship with both *Availability of housemaid* and *Family Support*.
- The correlation coefficient between *Availability of housemaid* and *Family Support* is *0.126,* which is a very weak positive relationship.
- The correlation between the same variables, such as *Continuity Decision* and *Continuity Decision*, *Business Type* and *Business Type, Availability of housemaid* and *Availability of housemaid*, & *Family Support* and *Family Support* is exactly *1.000*, which indicates that there is a perfect positive correlation between the variables.
- There is no correlation between the variables that is exactly 0, indicating that there is a relationship between all the variables.
- There is no correlation between the variables that is between -1 to 0, indicating that there is no negative relationship between the variables.

Pearson's r (*) method used in *Table 9* gives the correlation between profit earned by the online business owners and their time spent on business works.

- The correlation coefficient between *Profit* and *Business working-hours* is *0.114*, that is less than 0.2, means that there is a very weak positive relationship between both the variables.
- The correlation between the same variables, *Profit* and *Profit* & *Business working-hours* and *Business working-hours* is exactly *1.000,* indicating that there is a perfect positive correlation between the variables.

Table 10: Correlation between the Business Ratings

	Experience of business	***Customers***	***Responses from customers***	***Profit satisfaction***
Experience of business	1			
Customers	0.402	1		
Responses from customers	0.373	0.731	1	
Profit satisfaction	0.574	0.671	0.610	1

[*STATA command for Spearman's rho: *spearman experienceofbusiness customers responsesfromcustomers profitsatisfaction*]

In *Table 7* above, the ratings of online business owners on their business experience, customers received, responses from customers and their profit satisfaction was shown in percentage. In *Table 10,* the correlation between the ratings are highlighted.

- The correlation coefficient between *Experience of business* and *Customers* is *0.402,* which means that there is a moderate positive relationship between both the variables. Experiences of business are influenced by some other factors other than Customers.
- The correlation coefficient between *Experience of business* and *Responses from customers* is *0.373*, that is greater than 0.2, means that there is a weak positive relationship between both the variables and Experiences of business are influenced by some other factors other than Responses from customers.

- The correlation coefficient between *Experiences of business* and *Profit satisfaction* is *0.574*, that is greater than 0.4, means that there is a moderate positive relationship between both the variables and Experiences of business are influenced by some other factors other than Profit satisfaction.
- Besides, the correlation coefficient between *Customers* and *Responses from customers* is *0.731*, and that of *Customers* and *Profit satisfaction* is *0.671*. This indicates that the *Customers* received have a strong positive relationship with Responses from customers and a moderate positive relationship with Profit satisfaction.
- The correlation coefficient between *Responses from customers* and *Profit satisfaction* is *0.610,* which is a moderate positive relationship; Responses from customers and Profit satisfaction are influenced by some other factors too.
- The correlation between the same variables, such as *Experiences of business* and *Experiences of business*, *Customers* and *Customers, Responses from customers* and *Responses from customers*, & *Profit satisfaction* and *Profit satisfaction* is exactly *1.000*, which indicates that there is a perfect positive correlation between the variables.
- There is no correlation between the variables that is exactly 0, indicating that there is no weak relationship between the variables.
- There is no correlation between the variables that is between -1 to 0, indicating that there is no negative relationship between the variables.

7.3 Using Multivariate Analysis

Regression Analysis:

Taking the decision of continuing business as dependent variable and the three factors: Type of business, Availability of housemaid and Family's support as independent variables with error term ε, we can get the following online business women model,

$$\textbf{\textit{Continuity decision}} = \beta_1 + \beta_2\ \textbf{\textit{BusiType}} + \ \beta_3\ \textbf{\textit{HM}} + \ \beta_4\ \textbf{\textit{FamSup}} + \varepsilon_{Cd}$$

$$\textit{Continuity decision} = 0.743 + 0.056\ \textit{BusiType} + 0.335\ \textit{HM} + \ 0.028\ \textit{FamSup} + \varepsilon_{Cd}$$

[* STATA command for Regression: *reg continuitydecision busitype HM famsup*]

From the above regression model we can interpret that,

- Considering, type of business, availability of housemaid and Family's support has no effect on online business, (i.e. BusiType = HM = FamSup = 0), the decision of online women entrepreneurs to continue their business after COVID'19 pandemic will change by *0.743* units.
- Considering, availability of housemaid and family's support has no effect on online business, (i.e. HM = FamSup = 0), if entrepreneurs come up with 1 more type of business, the decision of online women entrepreneurs to continue their business after COVID'19 pandemic will change by *0.056* units.
- Considering, type of business and family's support has no effect on online business, (i.e. BusiType = FamSup = 0), if availability of housemaid in 1 businesswoman owner's house increases, the decision of online women entrepreneurs to continue their business after COVID'19 pandemic will change by *0.335* units.
- Considering, type of business and availability of housemaid has no effect on online business, (i.e.BusiType = HM = 0), if family member's support for 1 online businesswoman increases, the decision of online women entrepreneurs to continue their business after COVID'19 pandemic will change by *0.028* units.

Confidence Interval: 95% confidence interval for β_2 is -0.068 to 0.181, for β_3 is -0.099 to 0.769 and for β_4 is -0.265 to 0.323. This means that if 100 confidence intervals are constructed, 95 of them will contain β_2, β_3, β_4 and 5 of them will not.

R-squared: 14.50 % of the variation of the continuity decision can be explained by the variables: Type of business, Availability of housemaid and Family's support.

F-test statistics: $F_{calculated} = 1.16$ & $F_{critical} = 2.81$ (at 5% level of significance)

Since $F_{calculated} < F_{critical}$, the null hypothesis is not rejected and the model is not overall statistically significant. That means, the independent variable in my study does not influence the decision of continuing online business after COVID linearly.

Heteroscedasticity:

To find out if there is heteroscedasticity in my model, white test is used. [* STATA command for White Test: "*scalar LMwhite = e(r2)*e(N)*", "*display LMwhite*", and "*imtest,white*"]. The test is run to verify exactly whether the variance of the random disturbance varies from one variable to another utilizing the following hypotheses:

In my model, chi2(8): This is the $\chi^2_{calculated}$, that is 5.89

Prob > chi2: P-value of χ^2 is 0.6593, which is bigger than any level of significance (0.05, 0.1, 0.01) and so I do not reject the null hypothesis and conclude that there is no heteroscedasticity in the model. Again, for chi2(8), $\chi^2_{critical}$ = 15.5073 at 0.05 level of significance. Therefore, $\chi^2_{calculated} < \chi^2_{critical}$ = 5.89 < 15.5073 = No heteroscedasticity present in the model.

8. Discussion:

In the findings and analysis section, the data collected through my questionnaires are explained in detail using different analysis methods. Now, looking back to my research questions:

- *How are online businesses influencing new women entrepreneurs in Chittagong City during the COVID'19 pandemic?*
- *Is online business effective enough to continue after COVID'19?*

In this section, I will try to answer the questions, using the data collected.

8.1 Factors Influencing Online Business

As discussed before, from literature review I found that there are four factors that are influencing online women entrepreneurs to operate business online during the COVID'19: COVID'19 pandemic, type of business, availability of housemaid and family's support.

60% of my online business woman respondents started their business during COVID, and 28 of the women out of 30 started online business (*Table 8*). 50% of the women are conducting clothing & jewellery business (*Table 6*). 66% have househelp, who helps in household chores and 21 out of 33 are part-timers (*Table 8*). 64% get support from family (*Table 6*). Looking at the

percentages, I could connect my study with the existing studies in literature review. In addition to the existing literature, this paper looks into more detail.

Support from family comes in the case of marketing, taking care of the household, taking care of children and taking care of delivering the products (*Figure 3*). As the majority of women business owners get support from family, they get to spend less time in household work. 48% of the women business owners do not have to do any household work and 14% work sometimes (*Table 6*). In case of time spent in household chores, the largest percentage which is 20% spent less than 7 hours on household work (*Figure 6*). Thus, women can give more time to their business work; two majority groups, 24% of business owners spent 21 to 28 hours per week and another 24% spent 35 to 42 hours per week doing business work (*Figure 2*).

Looking into the monetary terms from online business; investment on average that online business owners spend is BDT 10,001 to BDT 20,000 (*Figure 5*) and the profit on average is BDT 10,001 to BDT 20,000 per month (*Figure 6*). However, profit does not have a strong relationship with business working-hours (*Figure 7*). The relation is a very weak positive relation which can be said that profit in business is influenced by some other factors, which is possible.

8.2 Factors Influencing Continuity Decision

Continuing online business after COVID seems to be an interesting question to the respondents. Some of them said that online business was started because they had free time during the COVID. When asked the question of, if they want to continue their online business after COVID (*Table 8*), 78% marked “Yes” and 14% marked “Maybe”. Among the respondents who marked “yes”, 10% entrepreneurs earned a profit of BDT 40,001 to BDT 50,000 per month and the majority (24%) earned a profit of BDT 10,001 to BDT 20,000 per month.

As already mentioned in Section 8.1, the factors that may influence online business, I tried to find a relation between the factors and the continuity decision in *Table 9*. However, there is no strong relation between the factors (type of business, availability of housemaid & family support) with continuity decisions, but I found a positive relation. Looking at the correlation coefficient, it can be seen that continuity decisions are influenced by some other factors other than factors mentioned in this study.

Moreover, I used regression analysis to find out the relationship between dependent variables (continuity decision) and independent variables (type of business, availability of

housemaid & family support) (Section 7.3). From regression analysis, it is found that my independent variables do not influence the continuity decisions linearly. However, a more rigorous Probit regression model or Logistic regression model could be used to show the relation of continuity decision and the influencing factors. Due to lack of knowledge on the model, it was not possible to use the model in this study. Future researchers are motivated to use the model to find out the relationship between the variables.

Thus it can be said that, although these factors can influence new women entrepreneurs to start their online business during the COVID, these factors do not influence the continuity decision in this study.

8.3 Possible to Continue After Covid

78% of the respondents want to continue their business after COVID - this leads to the question, if it is really possible to continue the business after COVID. It is seen that 84% of the respondents answered that it is 'possible' (*Figure 4*). As 60% of the respondents started their business during the COVID, it is necessary to know what my respondents think about the online business in comparison to traditional business, which is operated manually (*Table 6*). 68% of the respondents answered that online business is easier to handle and saves time, in comparison to traditional business. Among this 68% of respondents, there are respondents who started their business manually and shifted online during COVID. However, 32 % of respondents answered that online business is tougher to handle and is time consuming. As this question was an open-ended question, I looked at a few reasons about why online business is tougher to handle.

"In manual business, the customers can see the products on their own, but online I have to keep on sharing live videos on Facebook. My connection is slow and I am not so good in social media marketing"

"I think nothing is easy.. Either its online or offline business.. Yes this is right that I don't need to go out frequently for my business. But it's not easy manage the online customers"

Ratings collected on online business (*Table 7*) shows that -

- ➔ 54% said they have a good experience of online business.
- ➔ 52% said that they usually get customers.
- ➔ 50% responded that customers like their products.
- ➔ 42% responded that they are very satisfied with the profit they earn from online business.

I looked at the correlation between the business ratings (*Table 10*) and found that all the variables share a moderate positive relationship, i.e., for instance, the response of customers influences the number of customers, the number of customers influences the profit satisfaction, profit satisfaction influences the experience of business. However, these are not the only factors that are influencing each other, there are some more variables that can influence the variables that are not mentioned in this study.

9. Evaluation Criteria:

This study follows a *Cross-sectional Research Design.* The data collected in this study is at a single point in time; the questionnaire sent was filled-up once by each respondent and the data on the variables were collected simultaneously. The variation in this study is in respect of online women business owners at one point in time. No variables in this study are manipulated.

For evaluation of this social research, I will be showing 3 criteria:

Reliability: The result of this study is repeatable. The results of my study are consistent with the studies that are done in different countries, mentioned in the literature review. These results are consistent and stable as it contains data from variation of women entrepreneurs. Thus, this study is reliable.

Replication: Any new study that will be done with my research questions and questionnaire will obtain a consistent result. My procedure is explained in detail in my Methodology (Section 6) and my questionnaire is attached in the Appendix section. Thus, this study is replicable.

Validity:

This study has *Measurement Validity,* the measure used in this study is reliable. Any new researcher using the same measure even with a random sampling will get the same result.

However, in this study *Internal Validity* is not possible. Internal validity can be ensured only if the sample is selected through randomized assignment. The sampling method in this study is convenience sampling, which is a non-probability sampling method. Although, we can say the factors like business type, availability of housemaid and family support influences online business, but the causality is not actually proved. Moreover, this study does not have a treatment and control group, to compare the result. Thus the variables in my study do not have a causal relationship and so internal validity can not be ensured.

External Validity is also not possible for this study. External Validity can be ensured by selecting the evaluation sample that can be generalized to the population of all eligible units by using one of several variations of random sampling. For this study, my sampling frame is inadequate and the evaluation sample is not selected through randomized assignment. Thus my study can not generalize all online women business owners of Chittagong city and so external validity can not be ensured.

Ecological Validity is also not possible for this study. This study is conducted through answering questionnaires, that means, I never met the respondents personally and did not compare the answer with their everyday life situation, that is my self-completion questionnaire is disrupting the "natural habitat". This study can not be generalized to real-life settings and so ecological validity will be low.

10. Limitations:

This study has certain limitations:

- Non-probability sampling method used in this study is the most important weakness of this study. The convenience sampling method used to select participants decreased the opportunity of using the findings as a definitive finding because it can not be generalized.
- This study is restricted to an adequate sampling frame and sample size that is constrained to time and cost.
- The COVID'19 pandemic limited my data collection to within Chittagong city and thus I failed to address the online business owners from a broader perspective.

11. Recommendations:

However, findings of this research should be treated as an initial step at explaining the factors that influence online business in Chittagong city, Bangladesh and it is suggested that researchers come forward with some potential determinants that can be expanded in further research.

- Study with a larger sample size which is randomly selected and can generalize the whole population of online women business owners can be done to increase the external validity of the study.

- An adequate sampling frame will reduce sampling bias and increase the strength of the study.
- Rigorous regression models like logistic regression models can be used in future studies that can show my regression model is statistically significant.
- Rigorous analysis on the causality of the variables can be done to increase the internal validity of the study.
- *Randomized Control Trial*: Selection of treatment and control group can bring more interesting results. Randomized assignment of treatment and control group from evaluation sample will increase the internal validity of this research. Randomized selection of evaluation samples from different variations of eligible populations (here, online businesswomen) will increase the external validity of this study.

12. Conclusion

As discussed in this paper, online business has increased women employment in different countries; the same can be said for Bangladesh too. This paper showed a study on women entrepreneurs handling online business, looking for data on what influenced their online business and if they want to continue their business after COVID pandemic. Majority of new entrepreneurs started their online business during COVID, online business related to cloth and jewelleries are more prominent, availability of housemaid and support from family gives more free time to the women entrepreneurs, so that they can give time in their business work. Majority of the women entrepreneurs want to continue their business after COVID and it is possible to continue the online business after COVID because online businesses are easier to handle and save time. This study contains a variation of women entrepreneurs as it contains the data collected from married women entrepreneurs, single women entrepreneurs, divorcee women entrepreneurs and single parent women entrepreneurs. However, the correlations between my variables showed a weak positive relationship, which can be said that there are few other factors that are not included in my study. This study was an initial point to this topic and future researchers should take this study as a starting point and try to prove my model through more rigorous analysis. Also use of *Randomized Control Trial* in this study is suggested, which will be a helpful tool to ensure internal and external validity in this study.

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Appendix: Questionnaire

		CODE
Var0001	Are you operating an Online Business?	
	☐ Yes	1
	☐ No	2
Var0002	What type of Business do you have?	
	☐ Cloth & Jewelleries	1
	☐ Bakery	2
	☐ Henna	3
	☐ Cosmetics	4
	☐ Graphic Design	5
	☐ Organic Food	6
Var0003	When did you start your business?	
	☐ Before COVID'19	1
	☐ During the COVID'19	2
Var0004	How have you been operating your business from the start?	1 2 3
	☐ Started operating manually before COVID'19, shifted to online during COVID'19	
	☐ Started operating online before COVID'19 and continuing	
	☐ Started online business during COVID'19	
Var0005	Which Age group are you in?	
	☐ 16-25	1
	☐ 26-35	2
	☐ 36-45	3
	☐ 46-55	4
Var0006	Do you have to do household chores?	1 2 3
	☐ Yes	
	☐ No	
	☐ Sometimes	
Var0007	If yes, How much time do you spend on Household chores every week? [If No, please skip this question]	
	☐ less than 7 hours	1
	☐ 7 - 14 hours	2
	☐ 21 - 28 hours	3
	☐ 35 - 42 hours	4
	☐ 49 hours - 56 hours	5
Var0008	How much time do you spend on your business work every week?	
	☐ less than 7 hours	1
	☐ 7 - 14 hours	2
	☐ 21 - 28 hours	3
	☐ 35 - 42 hours	4
	☐ 49 hours - 56 hours	5

CODE

Var0009 Do you have a Housemaid? **1 2**

- ☐ Yes
- ☐ No

Var0010 If yes, Is the housemaid [If No, please skip this question]-- **1 2**

- ☐ Full-timer
- ☐ Part-timer

Var0011 What is your Marital Status?

- ☐ Married **1**
- ☐ Single **2**
- ☐ Divorcee **3**
- ☐ Single Parent **4**

Var0012 Does your Family help you in your business work or Household works? **1 2 3**

- ☐ Yes
- ☐ No
- ☐ Sometimes

Var0013 If Yes, how do they help? [If No, please skip this question] _____

- ☐ Delivering products
- ☐ Drive for you
- ☐ Help with handling customers
- ☐ Marketing
- ☐ Doing household chores
- ☐ Help children with studies
- ☐ Spent time with Children

Var0014 How is the experience of Online business? _____

	1	2	3	4	5	
Poor	◯	◯	◯	◯	◯	Excellent

Var0015 Do you get customers? _____

	1	2	3	4	5	
Rarely	◯	◯	◯	◯	◯	Frequently

Var0016 How are the responses from customers? _____

	1	2	3	4	5	
Customers dislike my Products	◯	◯	◯	◯	◯	Customers like my Products

CODE

Var0017 Are you satisfied with the profit from your business? _____

	1	2	3	4	5	
Not Satisfied	◯	◯	◯	◯	◯	Very satisfied

Var0018 How much have you invested to start the business?

- ☐ Less than BDT 10,000 — 1
- ☐ BDT 10,001 - BDT 20,000 — 2
- ☐ BDT 20,001 - BDT 30,000 — 3
- ☐ BDT 30,001 - BDT 40,000 — 4
- ☐ BDT 40,001 - BDT 50,000 — 5

Var0019 Approximately, how much profit do you earn every month from your business?

- ☐ Less than BDT 10,000 — 1
- ☐ BDT 10,001 - BDT 20,000 — 2
- ☐ BDT 20,001 - BDT 30,000 — 3
- ☐ BDT 30,001 - BDT 40,000 — 4
- ☐ BDT 40,001 - BDT 50,000 — 5

Var0020 From your experience, what do you think about Online business in comparison to Traditional business? [Is it easier or tougher to handle? Is it savingmore time or saving less time? Does it save cost or increase cost?] _____

_____

Var0021 Do you want to continue your online business after COVID'19? **1 2 3**

- ☐ Yes
- ☐ No
- ☐ Maybe

Var0022 Is it possible to continue online business after COVID'19? Will the profit and response from customers be the same? [Can you provide an example?] _____

_____